

Loyola Marymount University

CMSI 3300 Artificial Intelligence

Fall 2024

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Announcements

- Classwork 2
  - Due **this Friday, 9/13 @5 pm**
- Homework 1
  - Will be released after today's lecture
  - Due in ~3.5 weeks, Friday 10/4 @5 pm

This Lecture

- Quantifying Uncertainty
- Intro to Probability Theory
- Probabilistic Semantics
- Conditional Distribution

Last Lecture

- Theorem Proving (Proof by Contradiction)
- Example
  - KB
    1. $X \vee Z \vee Y$
    2. $\neg Z \vee W \vee X$
    3. $\neg X \vee W$
    4. $\neg W$
  - Query: $\alpha = \neg X \wedge Y$

$$
\begin{aligned}
(\alpha \wedge \beta) &\equiv (\beta \wedge \alpha) && \text{commutativity of } \wedge \\
(\alpha \vee \beta) &\equiv (\beta \vee \alpha) && \text{commutativity of } \vee \\
((\alpha \wedge \beta) \wedge \gamma) &\equiv (\alpha \wedge (\beta \wedge \gamma)) && \text{associativity of } \wedge \\
((\alpha \vee \beta) \vee \gamma) &\equiv (\alpha \vee (\beta \vee \gamma)) && \text{associativity of } \vee \\
\neg(\neg\alpha) &\equiv \alpha && \text{double-negation elimination} \\
(\alpha \Rightarrow \beta) &\equiv (\neg\beta \Rightarrow \neg\alpha) && \text{contraposition} \\
(\alpha \Rightarrow \beta) &\equiv (\neg\alpha \vee \beta) && \text{implication elimination} \\
(\alpha \Leftrightarrow \beta) &\equiv ((\alpha \Rightarrow \beta) \wedge (\beta \Rightarrow \alpha)) && \text{biconditional elimination} \\
\neg(\alpha \wedge \beta) &\equiv (\neg\alpha \vee \neg\beta) && \text{De Morgan} \\
\neg(\alpha \vee \beta) &\equiv (\neg\alpha \wedge \neg\beta) && \text{De Morgan} \\
(\alpha \wedge (\beta \vee \gamma)) &\equiv ((\alpha \wedge \beta) \vee (\alpha \wedge \gamma)) && \text{distributivity of } \wedge \text{ over } \vee \\
(\alpha \vee (\beta \wedge \gamma)) &\equiv ((\alpha \vee \beta) \wedge (\alpha \vee \gamma)) && \text{distributivity of } \vee \text{ over } \wedge
\end{aligned}
$$

Figure 7.11 Standard logical equivalences. The symbols α, β, and γ stand for arbitrary sentences of propositional logic.

Motivating Example



- **Automated Agent**: taxi!automated
- **Goal**: deliver a passenger on time
- **Plan**: leave home 90 minutes before the flight departs
- **KB** (propositional logic):

L=leave 90min before departure T= arrive in time

1. $\neg L \Rightarrow \neg T$
2. $\neg L$

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3. $\neg T$ (not arriving on time)

What is the problem with this?

Limitations of Propositional Logic

- The list of rules is incomplete
- Proposition is Boolean -> **cannot** represent **chance / uncertainty**
- The rule $(\neg L \Rightarrow \neg T)$ does not easily scale with exceptions (e.g., road condition, accidents, etc.)

Uncertainty

- **Uncertainty** about an environment arises in **imperfect information problems** wherein (for example) the agent has partial sensor information, ignorance of the rules of the environment, or the environment is fundamentally non-deterministic.
- How to address this?

Probabilistic Reasoning

- **Probabilistic reasoning** (or probabilistic inference) models uncertainty in the environment in a statistical representation that **allows the agent to act as best as it can with the information it has available**.
- The more our agents know about their environments, the less uncertainty.

Introduction to Probability Theory

- Propositional Logic
  - Truth Table -> Models
- Probability Theory
  - Truth Table -> ?

World	L	T
w_0	F	F
w_1	F	T
w_2	T	F
w_3	T	T

Intuition: Some worlds are **more likely** than others

Idea: Assign some **probability % to worlds** proportionate to likelihood of being witnessed

Probabilistic Logic Syntax

Propositional Entity	Probabilistic Analog	Probabilistic Syntax	Interpretation
Proposition	Event / Random Variable	**Capitalized** letters/words	- Need not be binary - Discrete/Continuous - E.g., L, T
(Truth) Values	(Event) Values	**Lowercase** letters/words	- The value that a random variable attains in a world. - E.g., T = “in time”

Probabilistic Logic Syntax

Propositional Entity	Probabilistic Analog	Probabilistic Syntax	Interpretation
Truth Table	Probability Distribution	$P(A \wedge B \wedge C \wedge \cdots) = P(A, B, C, \ldots)$	- Provides a **functional mapping** between variables and probability values that determines how likely each world is. - E.g., $P(L, T)$
Models	Probability Values	$P(A = a, B = b, \ldots) = P(a, b, \ldots)$	- Provides a **single probability value** for any instantiation of variables - E.g., $P(T = true) = 0.6 = 60\%$

Example – Discrete Variables

- For discrete variables, probability distributions are defined **tabularly**.
- **Joint Probability Table (JPT)**: the fully-specified table that provides this mapping for all variables in the model.
- Formally, $P(A, B, C, \dots) = P(A \wedge B \wedge C \wedge \cdots)$ define the likelihood of seeing all combos of variable values.

Example – Discrete Variables

World	L	T	$P(L,T)$
w_0	F	F	$P(L = false, T = false) = 0.3$
w_1	F	T	$P(L = false, T = true) = 0.1$
w_2	T	F	$P(L = true, T = false) = 0.1$
w_3	T	T	$P(L = true, T = true) = 0.5$

Note: JPT describe the **probability** of seeing the variable instantiations in each world **before** any other evidence.

Any characteristics of JPT?

Axiomatic Properties of Probability Theory

Rules that distributions must follow:

1. **Probability Value Range**: All probability values assigned to any given world w must range between 0 (impossible) and 1 (certain), inclusively: $0 \leq P(w) \leq 1$
2. **Sum of all probability values** must be 1: $\sum_w P(w) = 1$
3. **Law of Total Probability**: the probability of any subset of variables, α, is the sum of every possible world w that in that subset's models: $\mathrm{P}(\alpha) = \sum_{w \in M(\alpha)} P(w)$

All worlds w in models of α.

Example

- What is the raw probability that the passenger arrives in time, i.e., $P(T = true)$?

World	L	T	$P(L, T)$
w_0	F	F	$P(L = false, T = false) = 0.3$
w_1	F	T	$P(L = false, T = true) = 0.1$
w_2	T	F	$P(L = true, T = false) = 0.1$
w_3	T	T	$P(L = true, T = true) = 0.5$

$$P(T = true) = P(w_1) + P(w_3) = 0.1 + 0.5 = 0.6$$

Probabilistic Semantics

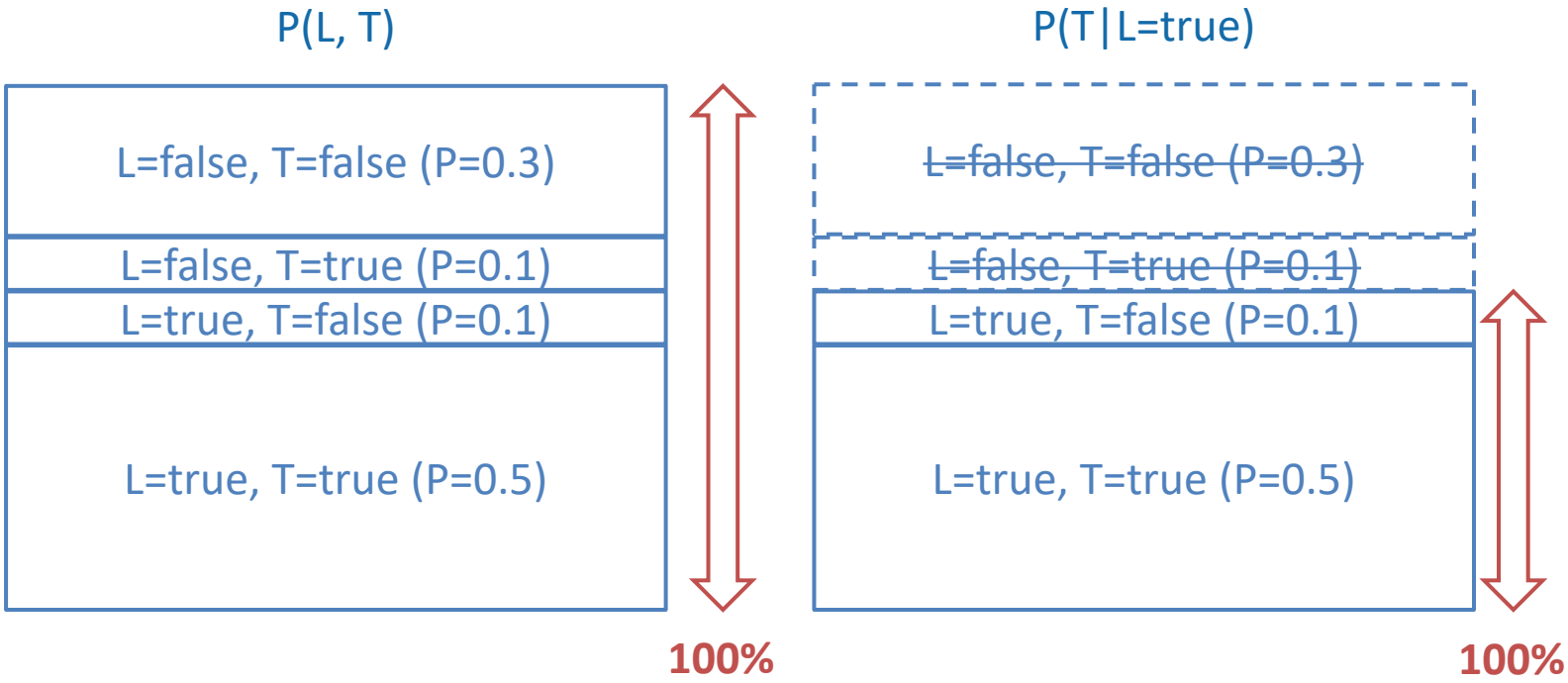
- Understand how **probabilities change** given **observed evidence**.
- Example:
  - Fact: the driver left 90 min before the flight departure time
  - Propositional logic: KB.tell(L)
  - Probabilistic logic:

World	L	T	$P(L,T)$	
~~w_0~~	~~F~~	~~F~~	~~$P(L = false, T = false) = 0.3$~~	0
~~w_1~~	~~F~~	~~T~~	~~$P(L = false, T = true) = 0.1$~~	0
w_2	T	F	$P(L = true, T = false) = 0.1$	/0.6
w_3	T	T	$P(L = true, T = true) = 0.5$	/0.6

No longer sum to 1?

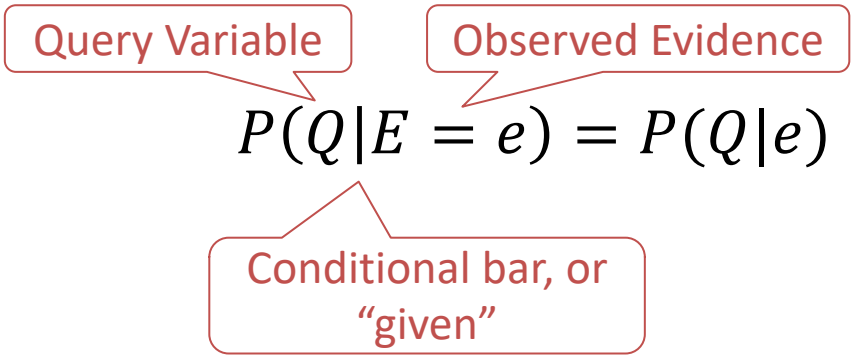
Conditional Distribution

World	L	T	$P(L,T)$
w_0	F	F	$P(L = false, T = false) = 0.3$
w_1	F	T	$P(L = false, T = true) = 0.1$
w_2	T	F	$P(L = true, T = false) = 0.1$
w_3	T	T	$P(L = true, T = true) = 0.5$



Conditional Distribution

- Normalizing joint values after accounting for observed evidence.
- Formally, a **conditional probability distribution** specifies the probability of some query variables Q given observed evidence E=e, is expressed as:



Conditional Distribution Table

World	L	T	$P(L,T)$	$P(T\|L)$
w_0	F	F	$P(L = false, T = false) = 0.3$	$\frac{P(w_0)}{P(L = false)} = \frac{0.3}{(0.3 + 0.1)}$
w_1	F	T	$P(L = false, T = true) = 0.1$	$\frac{P(w_1)}{P(L = false)} = \frac{0.1}{(0.3 + 0.1)}$
w_2	T	F	$P(L = true, T = false) = 0.1$	$\frac{P(w_2)}{P(L = true)} = \frac{0.1}{(0.1 + 0.5)}$
w_3	T	T	$P(L = true, T = true) = 0.5$	$\frac{P(w_3)}{P(L = true)} = \frac{0.5}{(0.1 + 0.5)}$

=1

=1

Conditioning Rule

- For any two sets of queries and evidence, we have

$$P(Q|e) = \frac{P(Q,e)}{P(e)} = \frac{joint\ probability\ of\ \{Q,e\}}{probability\ of\ \{e\}}$$

- Example

Assume the driver wants to guarantee that the passenger can arrive in time, should the driver accept the trip is the taxi cannot leave 90 min before the flight departure, i.e., L=false?

World	L	T	$P(L,T)$
w_0	F	F	$P(L = false, T = false) = 0.3$
w_1	F	T	$P(L = false, T = true) = 0.1$
w_2	T	F	$P(L = true, T = false) = 0.1$
w_3	T	T	$P(L = true, T = true) = 0.5$

No!

$$P(T = true|L = false) = \frac{P(L = false, T = true)}{P(L = false)} = \frac{P(w_1)}{P(w_0) + P(w_1)} = \frac{0.1}{0.3 + 0.1} = 0.25$$

Practice

- Given that $P(A=1|B=1)=0.4$ and $P(B=1)=0.2$, find $P(A=1,B=1)$ using the conditioning rule.

Practice

- Given the following joint over C,D, find P(C=1) using the law of total probability (LoTP). Show your work and all symbolic steps before plugging in numbers

P(C,D)	D=0	D=1
C=0	0.3	0.2
C=1	0.4	0.1